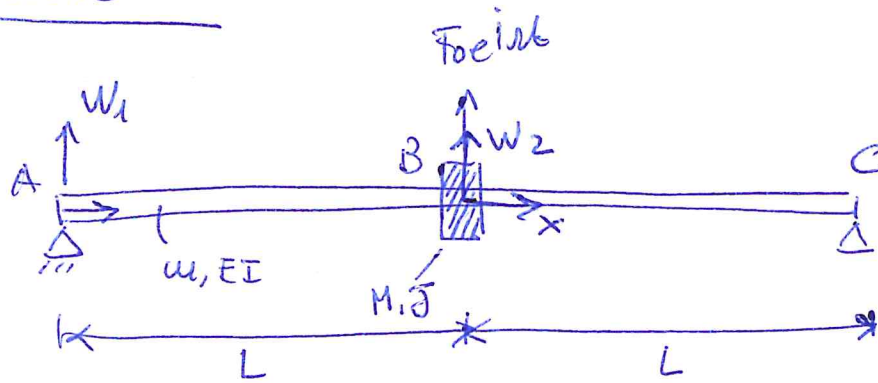
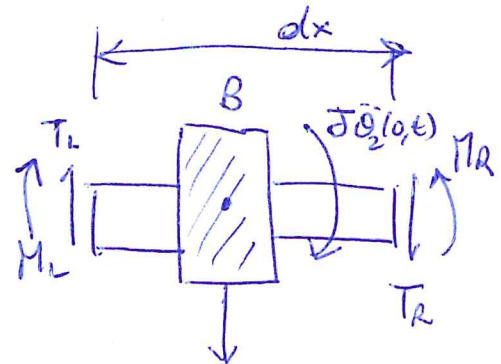
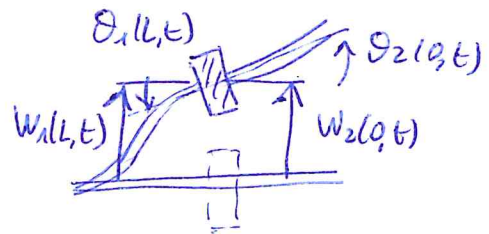


Written Test July 05th 2022

PROBLEM 2



- 1) $w_1(0, t) = 0$
- 2) $M_1(0, t) = 0$
- 3) $w_1(L, t) - w_2(0, t) = 0$
- 4) $\theta_1(L, t) - \theta_2(0, t) = 0$
- 5) $M_L + J\ddot{\theta}_2(0, t) - M_L + \cancel{T_L \frac{dx}{2}} + \cancel{T_R \frac{dx}{2}} = 0$
 $M_1(L, t) + J\ddot{\theta}_2(0, t) - M_2(0, t) = 0$
- 6) $\cancel{T_L} - M\ddot{w}_2(0, t) - \cancel{m dx \ddot{w}_2(0, t)} - \cancel{T_R} = 0$
 $T_1(L, t) - M\ddot{w}_2(0, t) - T_2(0, t) = 0$
- 7) $w_2(L, t) = 0$
- 8) $M_2(L, t) = 0$



$$F_{in} = M\ddot{w}_2(0, t) + m dx \ddot{w}_2(0, t)$$

$$w_2 = [A_2 \cos \gamma x + B_2 \sinh \gamma x + C_2 \cosh \gamma x + D_2 \sin \gamma x] \cos(\omega t + \varphi)$$

$$\ddot{w}_2(x, t) = -\omega^2 [A_2 \cos \gamma x + B_2 \sinh \gamma x + C_2 \cosh \gamma x + D_2 \sin \gamma x] \cos(\omega t + \varphi)$$

$$\ddot{\theta}_2(x, t) = -\omega^2 \gamma [-A_2 \sinh \gamma x + B_2 \cos \gamma x + C_2 \sin \gamma x + D_2 \cosh \gamma x] \cos(\omega t + \varphi)$$

$$A_1 + C_1 = 0$$

$$\cancel{\gamma^2 EI} (-A_1 + C_1) = 0$$

$$A_1 \cos \gamma L + B_1 \sinh \gamma L + C_1 \cosh \gamma L + D_1 \sinh \gamma L - A_2 - C_2 = 0$$

$$\cancel{\gamma} (-A_1 \sinh \gamma L + B_1 \cos \gamma L + C_1 \sinh \gamma L + D_1 \cosh \gamma L) - \cancel{\gamma} (B_2 + D_2) = 0$$

$$\cancel{\gamma^2 EI} (-A_1 \cos \gamma L - B_1 \sinh \gamma L + C_1 \cosh \gamma L + D_1 \sinh \gamma L) - \cancel{\gamma^2 \frac{EI}{m}} \gamma (B_2 + D_2) +$$

$$- \cancel{\gamma^2 EI} (-A_2 + C_2) = 0$$

$$\gamma^3 EI (A_1 \sinh \gamma L - B_1 \cos \gamma L + C_1 \sinh \gamma L - D_1 \cosh \gamma L) + \gamma^4 \frac{EI}{m} M (A_2 + C_2) +$$

$$- \cancel{\gamma^3 EI} (-B_2 + D_2) = 0$$

$$A_2 \cos \gamma L + B_2 \sinh \gamma L + C_2 \cosh \gamma L + D_2 \sinh \gamma L = 0$$

$$\cancel{\gamma^2 EI} (-A_2 \cos \gamma L - B_2 \sinh \gamma L + C_2 \cosh \gamma L + D_2 \sinh \gamma L) = 0$$

$$\underline{x} = \{ A_1 \ B_1 \ C_1 \ D_1 \ A_2 \ B_2 \ C_2 \ D_2 \}^T$$

$$[H] = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \cos \gamma L & \sinh \gamma L & \cosh \gamma L & \sinh \gamma L & -1 & 0 & -1 & 0 \\ -\sinh \gamma L & \cos \gamma L & \sinh \gamma L & \cosh \gamma L & 0 & -1 & 0 & -1 \\ -\cos \gamma L & -\sinh \gamma L & \cosh \gamma L & \sinh \gamma L & 1 & -\gamma^3 \gamma / m & -1 & -\gamma^3 \gamma / m \\ \sinh \gamma L & -\cos \gamma L & \sinh \gamma L & \cosh \gamma L & \frac{\gamma M}{m} & 1 & \frac{\gamma M}{m} & -1 \\ 0 & 0 & 0 & 0 & \cos \gamma L & \sinh \gamma L & \cosh \gamma L & \sinh \gamma L \\ 0 & 0 & 0 & 0 & -\cos \gamma L & -\sinh \gamma L & \cosh \gamma L & \sinh \gamma L \end{bmatrix}$$

$$K_{ui} = K_{ui, \text{beam}}$$

$$W_{ui} = W_{ui, \text{beam}} + M \phi_i^2(L) + \gamma \phi_i'^2(L)$$

ϕ_i is the i -th row of the computed by the Matlab's function "modeshape"

matrix "w"